

Public Policy for the Poor? A Randomized Evaluation of the Mexican Universal Health Insurance Program

Gary King
Institute for Quantitative Social Science
Harvard University

Joint work with Emmanuela Gakidou, Kosuke Imai, Jason Lakin, Ryan T. Moore, Clayon Nall, Nirmala Ravishankar, Manett Vargas, Martha María Téllez-Rojo, Juan Eugenio Hernández Ávila, Mauricio Hernández Ávila, Héctor Hernández Llamas

(Talk at Harvard School of Public Health, 5/1/2012)

Project References

Project References

- Gary King et al., A 'Politically Robust' Experimental Design for Public Policy Evaluation, with Application to the Mexican Universal Health Insurance Program *Journal of Policy Analysis and Management*, 26, 3 (2007): 479-506.

Project References

- Gary King et al., A 'Politically Robust' Experimental Design for Public Policy Evaluation, with Application to the Mexican Universal Health Insurance Program *Journal of Policy Analysis and Management*, 26, 3 (2007): 479-506.
- Kosuke Imai, Gary King, and Clayton Nall. The Essential Role of Pair Matching in Cluster-Randomized Experiments, with Application to the Mexican Universal Health Insurance Evaluation *Statistical Science*, 24, 1 (2009): pp. 29–53.

Project References

- Gary King et al., A 'Politically Robust' Experimental Design for Public Policy Evaluation, with Application to the Mexican Universal Health Insurance Program *Journal of Policy Analysis and Management*, 26, 3 (2007): 479-506.
- Kosuke Imai, Gary King, and Clayton Nall. The Essential Role of Pair Matching in Cluster-Randomized Experiments, with Application to the Mexican Universal Health Insurance Evaluation *Statistical Science*, 24, 1 (2009): pp. 29–53.
- Gary King et al., Public Policy for the Poor? A Randomized 10-Month Evaluation of the Mexican Universal Health Insurance Program *The Lancet*, 373 (25 April 2009): 1447-1454.

Project References

- Gary King et al., A 'Politically Robust' Experimental Design for Public Policy Evaluation, with Application to the Mexican Universal Health Insurance Program *Journal of Policy Analysis and Management*, 26, 3 (2007): 479-506.
- Kosuke Imai, Gary King, and Clayton Nall. The Essential Role of Pair Matching in Cluster-Randomized Experiments, with Application to the Mexican Universal Health Insurance Evaluation *Statistical Science*, 24, 1 (2009): pp. 29–53.
- Gary King et al., Public Policy for the Poor? A Randomized 10-Month Evaluation of the Mexican Universal Health Insurance Program *The Lancet*, 373 (25 April 2009): 1447-1454.
- <http://gking.harvard.edu/projects/mex.shtml>

Seguro Popular: A Massive Reform

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.
- Mexican Health Policy: centralized $\rightsquigarrow$ decentralized $\rightsquigarrow$ **stewardship**

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.
- Mexican Health Policy: centralized $\rightsquigarrow$ decentralized $\rightsquigarrow$ **stewardship**
- Cost in 2005: \$795.5 million in new money

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.
- Mexican Health Policy: centralized $\rightsquigarrow$ decentralized $\rightsquigarrow$ **stewardship**
- Cost in 2005: \$795.5 million in new money
- Cost when fully implemented: additional 1% of GDP

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.
- Mexican Health Policy: centralized $\rightsquigarrow$ decentralized $\rightsquigarrow$ **stewardship**
- Cost in 2005: \$795.5 million in new money
- Cost when fully implemented: additional 1% of GDP
- One of the largest health reforms of any country in last 2 decades

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.
- Mexican Health Policy: centralized $\rightsquigarrow$ decentralized $\rightsquigarrow$ **stewardship**
- Cost in 2005: \$795.5 million in new money
- Cost when fully implemented: additional 1% of GDP
- One of the largest health reforms of any country in last 2 decades
- Most visible accomplishment of the Fox administration

Seguro Popular: A Massive Reform

- medical services, preventive care, pharmaceuticals, and financial health protection
- beneficiaries: 50M Mexicans (half of the population) with no regular access to health care, particularly those with low incomes.
- Mexican Health Policy: centralized $\rightsquigarrow$ decentralized $\rightsquigarrow$ **stewardship**
- Cost in 2005: \$795.5 million in new money
- Cost when fully implemented: additional 1% of GDP
- One of the largest health reforms of any country in last 2 decades
- Most visible accomplishment of the Fox administration
- Major issue in the 2006 presidential campaign

Goals of SP & Evaluation Outcome Measures

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular
- Health Care Facilities

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular
- Health Care Facilities
 - Operations, office visits, emergencies, personnel, infrastructure and equipment, drug inventory.

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular
- Health Care Facilities
 - Operations, office visits, emergencies, personnel, infrastructure and equipment, drug inventory.
- Health

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular
- Health Care Facilities
 - Operations, office visits, emergencies, personnel, infrastructure and equipment, drug inventory.
- Health
 - Health status

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular
- Health Care Facilities
 - Operations, office visits, emergencies, personnel, infrastructure and equipment, drug inventory.
- Health
 - Health status
 - All-cause mortality

Goals of SP & Evaluation Outcome Measures

- Financial Protection (money for the poor rarely makes it there)
 - Out-of-pocket expenditure
 - Catastrophic expenditure (8.4% of households, & 10% of the poor, spend > 30% of annual disposable income on health)
 - Impoverishment due to health care payments
- Health System Effective Coverage
 - Percent of population receiving appropriate treatment by disease
 - Responsiveness of Seguro Popular
 - Satisfaction of affiliates with Seguro Popular
- Health Care Facilities
 - Operations, office visits, emergencies, personnel, infrastructure and equipment, drug inventory.
- Health
 - Health status
 - All-cause mortality
 - Cause-specific mortality

SP Evaluation

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)
 - Like in science: make themselves vulnerable to being proven wrong

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)
 - Like in science: make themselves vulnerable to being proven wrong
 - If we show SP is a success: elimination would be difficult

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)
 - Like in science: make themselves vulnerable to being proven wrong
 - If we show SP is a success: elimination would be difficult
 - If SP is a failure: who cares about extending it

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)
 - Like in science: make themselves vulnerable to being proven wrong
 - If we show SP is a success: elimination would be difficult
 - If SP is a failure: who cares about extending it
- The largest randomized health policy experiment in history

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)
 - Like in science: make themselves vulnerable to being proven wrong
 - If we show SP is a success: elimination would be difficult
 - If SP is a failure: who cares about extending it
- The largest randomized health policy experiment in history
- One of the largest policy experiments to date

- Frenk and Fox asked: How can one democratically elected government “tie the hands” of their successors?
 - Commission an independent evaluation
 - (They are true believers in SP)
 - Like in science: make themselves vulnerable to being proven wrong
 - If we show SP is a success: elimination would be difficult
 - If SP is a failure: who cares about extending it
- The largest randomized health policy experiment in history
- One of the largest policy experiments to date
- First cohort: 148 geographic areas, 1,380 localities, $\approx 118,569$ households, and $\approx 534,457$ people

Lessons from Previous Public Policy Experiments

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?
 - all perfectly legitimate; a natural consequence in a democracy

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?
 - all perfectly legitimate; a natural consequence in a democracy
- E.g., Oportunidades program: Some governors “miraculously” found money for control groups to participate too (numerous similar examples worldwide)

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?
 - all perfectly legitimate; a natural consequence in a democracy
- E.g., Oportunidades program: Some governors “miraculously” found money for control groups to participate too (numerous similar examples worldwide)
- Previous evaluation designs ignored democratic politics

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?
 - all perfectly legitimate; a natural consequence in a democracy
- E.g., Oportunidades program: Some governors “miraculously” found money for control groups to participate too (numerous similar examples worldwide)
- Previous evaluation designs ignored democratic politics
- We developed a new research design & new methods for Mexico:

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?
 - all perfectly legitimate; a natural consequence in a democracy
- E.g., Oportunidades program: Some governors “miraculously” found money for control groups to participate too (numerous similar examples worldwide)
- Previous evaluation designs ignored democratic politics
- We developed a new research design & new methods for Mexico:
 - includes **fail-safe components** for when politics intervenes

Lessons from Previous Public Policy Experiments

- Most large scale public policy experiments fail
- Many failures are political
 - politicians: need to pursue short term goals
 - citizens: you plan to *randomly* assign *me*?
 - all perfectly legitimate; a natural consequence in a democracy
- E.g., Oportunidades program: Some governors “miraculously” found money for control groups to participate too (numerous similar examples worldwide)
- Previous evaluation designs ignored democratic politics
- We developed a new research design & new methods for Mexico:
 - includes **fail-safe components** for when politics intervenes
 - uses data far more efficiently to **find effects** and **save money**

Example of Fail-Safe Design Procedure (CR vs. MPR)

Example of Fail-Safe Design Procedure (CR vs. MPR)

- 1 Complete Randomization (used in Oportunidades evaluation)

Example of Fail-Safe Design Procedure (CR vs. MPR)

- 1 **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area

Example of Fail-Safe Design Procedure (CR vs. MPR)

- 1 **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:

Example of Fail-Safe Design Procedure (CR vs. MPR)

- 1 **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

Example of Fail-Safe Design Procedure (CR vs. MPR)

- ① **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone
- ② **Matched-Pair Randomization** (used in **Seguro Popular** evaluation)

Example of Fail-Safe Design Procedure (CR vs. MPR)

- ① **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone
- ② **Matched-Pair Randomization** (used in **Seguro Popular** evaluation)
 - Match areas in pairs on background characteristics

Example of Fail-Safe Design Procedure (CR vs. MPR)

- ① **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone
- ② **Matched-Pair Randomization** (used in **Seguro Popular** evaluation)
 - Match areas in pairs on background characteristics
 - Flip coin once for each pair: one area within each pair gets the program

Example of Fail-Safe Design Procedure (CR vs. MPR)

- ① **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone
- ② **Matched-Pair Randomization** (used in **Seguro Popular** evaluation)
 - Match areas in pairs on background characteristics
 - Flip coin once for each pair: one area within each pair gets the program
 - If one area is lost:

Example of Fail-Safe Design Procedure (CR vs. MPR)

- ① **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone
- ② **Matched-Pair Randomization** (used in **Seguro Popular** evaluation)
 - Match areas in pairs on background characteristics
 - Flip coin once for each pair: one area within each pair gets the program
 - If one area is lost:
 - Drop the other member of the pair

Example of Fail-Safe Design Procedure (CR vs. MPR)

- ① **Complete Randomization** (used in **Oportunidades** evaluation)
 - Flip coin to assign program to each area
 - If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone
- ② **Matched-Pair Randomization** (used in **Seguro Popular** evaluation)
 - Match areas in pairs on background characteristics
 - Flip coin once for each pair: one area within each pair gets the program
 - If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

② Matched-Pair Randomization (used in Seguro Popular evaluation)

- Match areas in pairs on background characteristics
- Flip coin once for each pair: one area within each pair gets the program
- If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept
 - Treated and control groups are still protected by randomization: advantages of the experiment survives

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

② Matched-Pair Randomization (used in Seguro Popular evaluation)

- Match areas in pairs on background characteristics
- Flip coin once for each pair: one area within each pair gets the program
- If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept
 - Treated and control groups are still protected by randomization: advantages of the experiment survives
- With our new statistical methods, the design:

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

② Matched-Pair Randomization (used in Seguro Popular evaluation)

- Match areas in pairs on background characteristics
- Flip coin once for each pair: one area within each pair gets the program
- If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept
 - Treated and control groups are still protected by randomization: advantages of the experiment survives
- With our new statistical methods, the design:
 - **More efficient:** up to 38 times!

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

② Matched-Pair Randomization (used in Seguro Popular evaluation)

- Match areas in pairs on background characteristics
- Flip coin once for each pair: one area within each pair gets the program
- If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept
 - Treated and control groups are still protected by randomization: advantages of the experiment survives
- With our new statistical methods, the design:
 - **More efficient:** up to 38 times!
 - **Smaller standard errors:** up to 6 times smaller

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

② Matched-Pair Randomization (used in Seguro Popular evaluation)

- Match areas in pairs on background characteristics
- Flip coin once for each pair: one area within each pair gets the program
- If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept
 - Treated and control groups are still protected by randomization: advantages of the experiment survives
- With our new statistical methods, the design:
 - **More efficient:** up to 38 times!
 - **Smaller standard errors:** up to 6 times smaller
 - **We can find effects where complete randomization cannot**

Example of Fail-Safe Design Procedure (CR vs. MPR)

① Complete Randomization (used in Oportunidades evaluation)

- Flip coin to assign program to each area
- If one area is lost:
 - treated and control groups are incomparable
 - all advantages of randomization are gone

② Matched-Pair Randomization (used in Seguro Popular evaluation)

- Match areas in pairs on background characteristics
- Flip coin once for each pair: one area within each pair gets the program
- If one area is lost:
 - Drop the other member of the pair
 - Remaining pairs are kept
 - Treated and control groups are still protected by randomization: advantages of the experiment survives
- With our new statistical methods, the design:
 - **More efficient:** up to 38 times!
 - **Smaller standard errors:** up to 6 times smaller
 - **We can find effects where complete randomization cannot**
 - **Far less expensive** for the same impact

Detailed Design Summary

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico's 31 states; each includes a health clinic and catchment area

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)
- 3 Match clusters in pairs on background characteristics.

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)
- 3 Match clusters in pairs on background characteristics.
- 4 Select 74 pairs (based on necessary political criteria, closeness of the match, likelihood of compliance)

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)
- 3 Match clusters in pairs on background characteristics.
- 4 Select 74 pairs (based on necessary political criteria, closeness of the match, likelihood of compliance)
- 5 Randomly assign one in each pair to receive encouragement to affiliate, better health facilities, drugs, and doctors

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)
- 3 Match clusters in pairs on background characteristics.
- 4 Select 74 pairs (based on necessary political criteria, closeness of the match, likelihood of compliance)
- 5 Randomly assign one in each pair to receive encouragement to affiliate, better health facilities, drugs, and doctors
- 6 Conduct baseline survey of each cluster’s health facility

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)
- 3 Match clusters in pairs on background characteristics.
- 4 Select 74 pairs (based on necessary political criteria, closeness of the match, likelihood of compliance)
- 5 Randomly assign one in each pair to receive encouragement to affiliate, better health facilities, drugs, and doctors
- 6 Conduct baseline survey of each cluster’s health facility
- 7 Survey $\approx 32,000$ random households in 50 of the 74 treated and control unit pairs (chosen based on likelihood of compliance with encouragement and similarity of the clusters within pair)

Detailed Design Summary

- 1 Define 12,284 “health clusters” that tile Mexico’s 31 states; each includes a health clinic and catchment area
- 2 Persuaded 13 of 31 states to participate (7,078 clusters)
- 3 Match clusters in pairs on background characteristics.
- 4 Select 74 pairs (based on necessary political criteria, closeness of the match, likelihood of compliance)
- 5 Randomly assign one in each pair to receive encouragement to affiliate, better health facilities, drugs, and doctors
- 6 Conduct baseline survey of each cluster’s health facility
- 7 Survey $\approx 32,000$ random households in 50 of the 74 treated and control unit pairs (chosen based on likelihood of compliance with encouragement and similarity of the clusters within pair)
- 8 Repeat surveys in 10 months and subsequently to see effects

Matched-Pair Cluster-Randomized Designs in Polisci

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design:

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary:

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $MPDs \geq \text{Complete Randomization}$ w.r.t.:

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $MPDs \geq \text{Complete Randomization}$ w.r.t.: efficiency,

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $MPDs \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias,

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $MPDs \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power,

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $MPDs \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power, estimator simplicity,

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $MPDs \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power, estimator simplicity, and robustness to political intervention

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $\text{MPDs} \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power, estimator simplicity, and robustness to political intervention
- Proportion of previous CREs in polisci that use MPDs:

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $\text{MPDs} \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power, estimator simplicity, and robustness to political intervention
- Proportion of previous CREs in polisci that use MPDs: 0%

Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $\text{MPDs} \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power, estimator simplicity, and robustness to political intervention
- Proportion of previous CREs in polisci that use MPDs: 0%
- Conclusion: we're leaving a lot of information on the table!

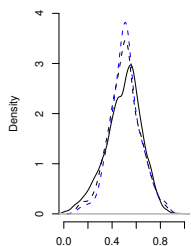
Matched-Pair Cluster-Randomized Designs in Polisci

- Special research designs require special methods
- Prop. of polisci CREs which ignore the design: 100%
- Prop. of polisci CREs making more assumptions than necessary: 100%
- $\text{MPDs} \geq \text{Complete Randomization}$ w.r.t.: efficiency, bias, power, estimator simplicity, and robustness to political intervention
- Proportion of previous CREs in polisci that use MPDs: 0%
- Conclusion: we're leaving a lot of information on the table!
- Imai-King-Nall: prove above results and offer simple estimators for MPDs making minimal assumptions for both **intent to treat** and **complier average treatment** effects

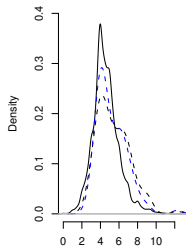
Remaining in study: 148 clusters (74 pairs) in 7 states



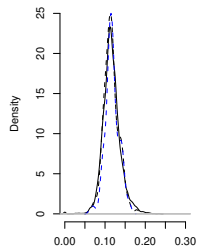
Clusters are Representative On Measured Variables



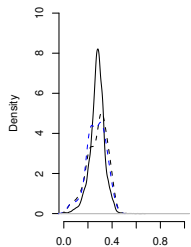
Prop earning <2 min wages



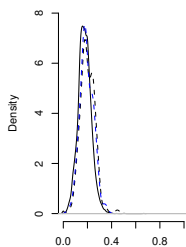
Mean Years Education



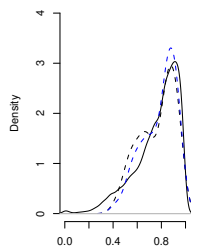
Prop aged 0-4 years old



Prop Employed

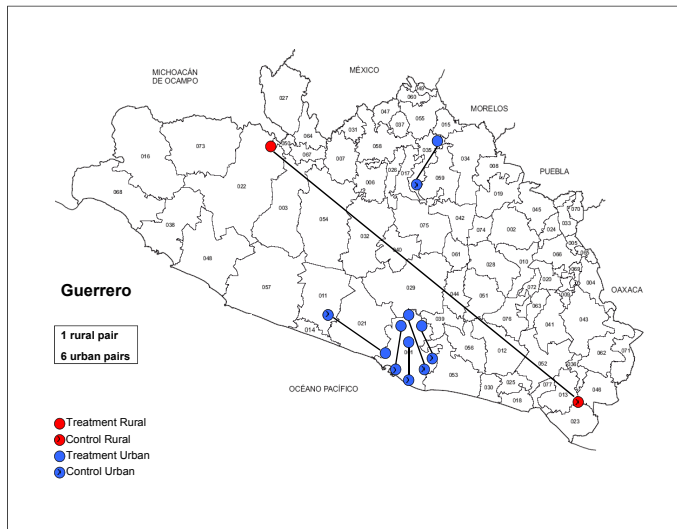


Prop Female-headed HH



Prop w/o Soc Sec Rights

Matched Pairs, Guerrero

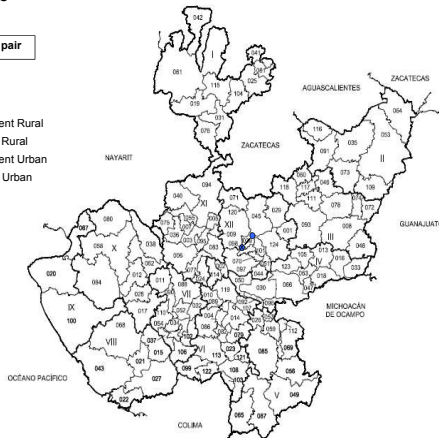


Matched Pairs, Jalisco

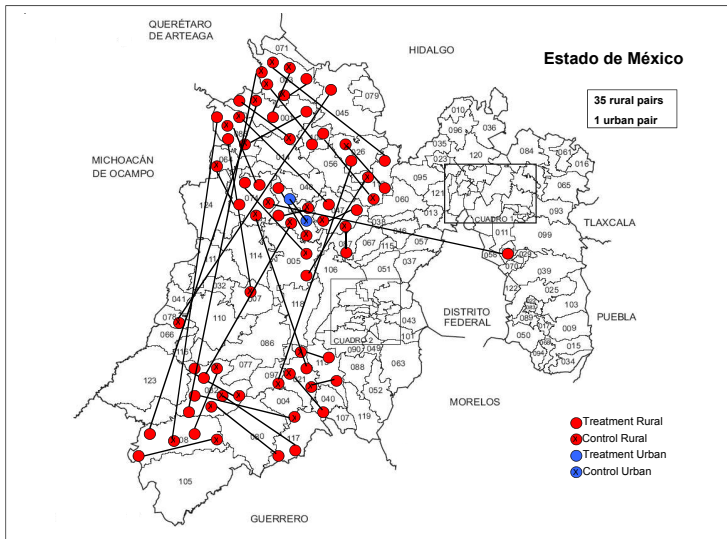
Jalisco

1 urban pair

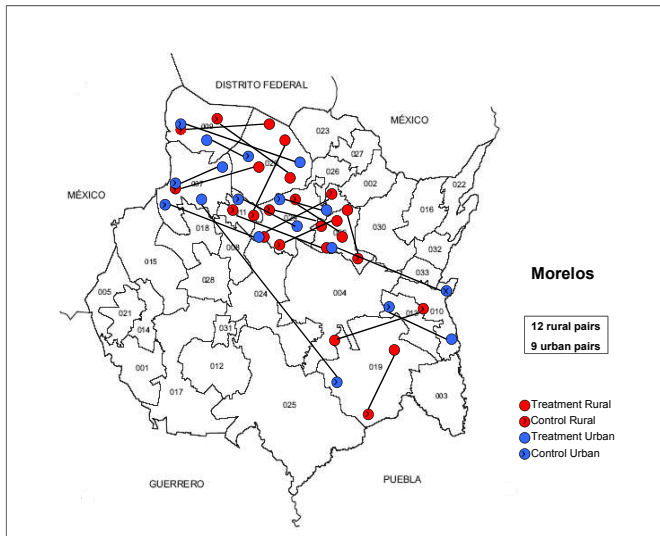
- Treatment Rural
- Control Rural
- Treatment Urban
- Control Urban



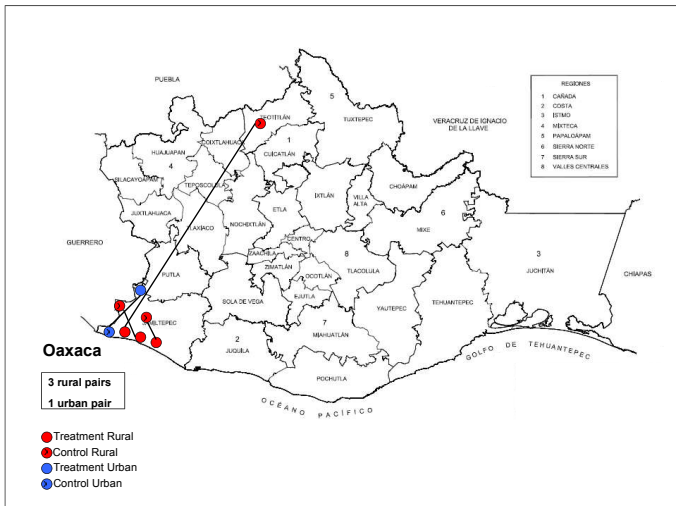
Matched Pairs, Estado de México



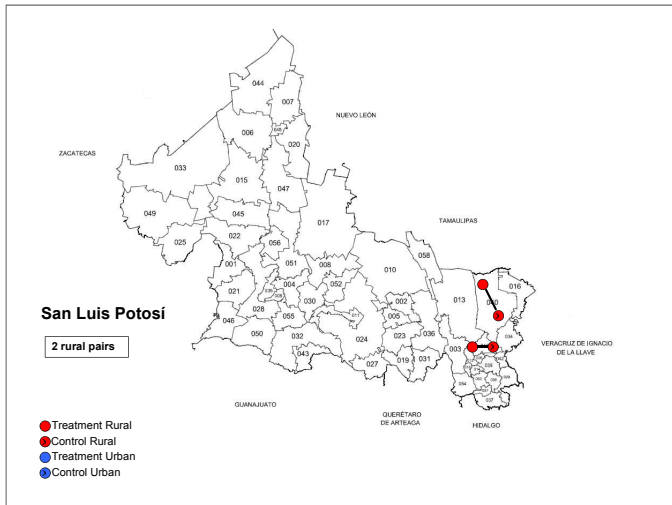
Matched Pairs, Morelos



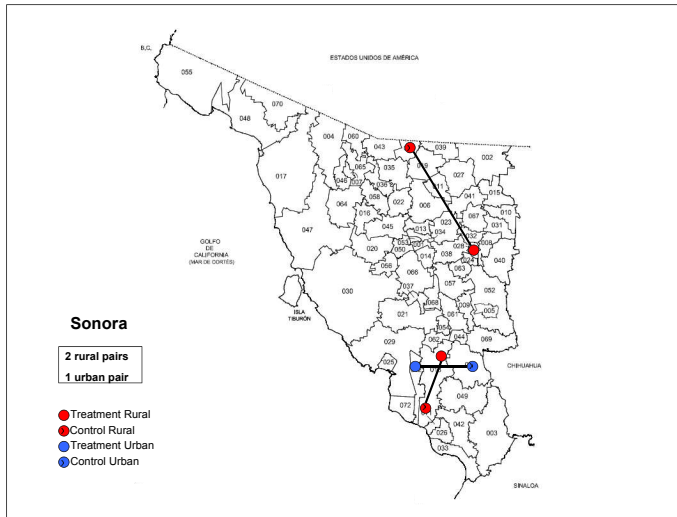
Matched Pairs, Oaxaca



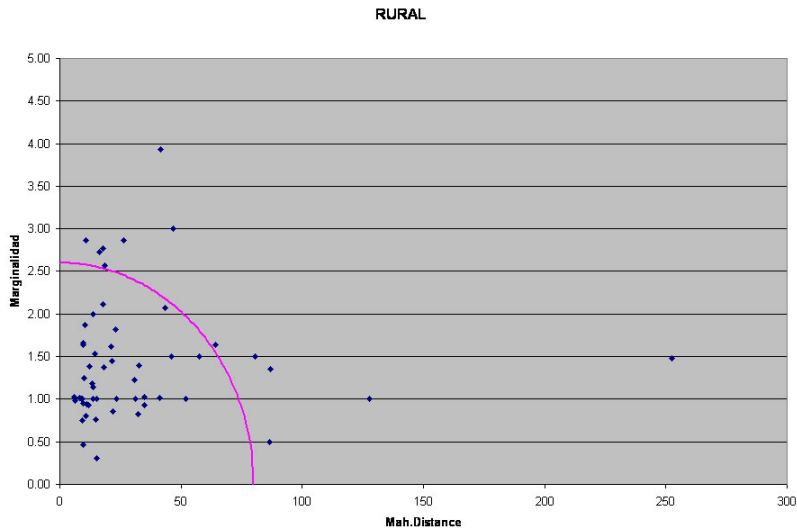
Matched Pairs, San Luis Potosí



Matched Pairs, Sonora



Choosing Pairs for the Survey



Design and Analysis Strategy is Triply Robust

Design and Analysis Strategy is Triply Robust

Design has three parts

Design and Analysis Strategy is Triply Robust

Design has three parts

- 1 Matching pairs on observed covariates

Design and Analysis Strategy is Triply Robust

Design has three parts

- 1 Matching pairs on observed covariates
- 2 Randomization of treatment within pairs

Design and Analysis Strategy is Triply Robust

Design has three parts

- ① Matching pairs on observed covariates
- ② Randomization of treatment within pairs
- ③ If necessary statistically adjust for differences

Design and Analysis Strategy is Triply Robust

Design has three parts

- 1 Matching pairs on observed covariates
- 2 Randomization of treatment within pairs
- 3 If necessary statistically adjust for differences

Triple Robustness

If matching **or** randomization **or** statistical analysis is right, but the other two are wrong, results are still unbiased

Design and Analysis Strategy is Triply Robust

Design has three parts

- 1 Matching pairs on observed covariates
- 2 Randomization of treatment within pairs
- 3 If necessary statistically adjust for differences

Triple Robustness

If matching **or** randomization **or** statistical analysis is right, but the other two are wrong, results are still unbiased

Two Additional Checks if Triple Robustness Fails

Design and Analysis Strategy is Triply Robust

Design has three parts

- 1 Matching pairs on observed covariates
- 2 Randomization of treatment within pairs
- 3 If necessary statistically adjust for differences

Triple Robustness

If matching **or** randomization **or** statistical analysis is right, but the other two are wrong, results are still unbiased

Two Additional Checks if Triple Robustness Fails

- 1 If one of the three works, then “effect of SP” on time 0 outcomes (measured in baseline survey) must be zero

Design and Analysis Strategy is Triply Robust

Design has three parts

- 1 Matching pairs on observed covariates
- 2 Randomization of treatment within pairs
- 3 If necessary statistically adjust for differences

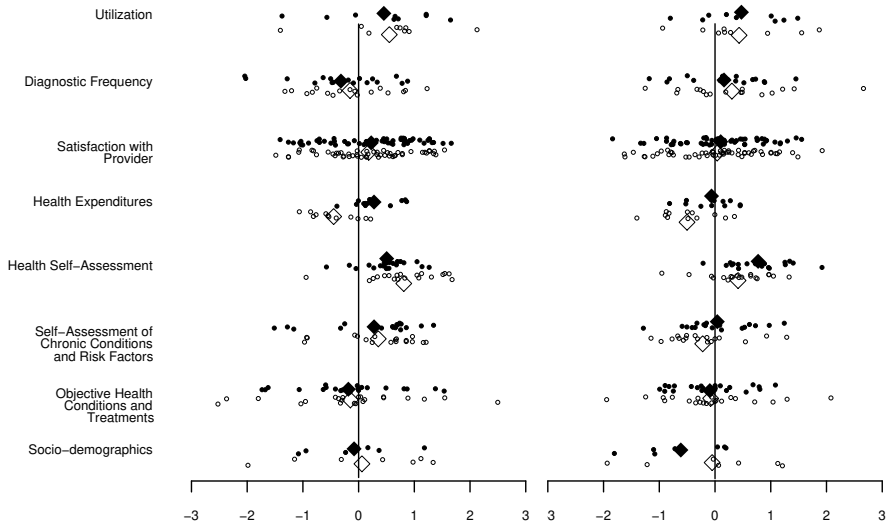
Triple Robustness

If matching **or** randomization **or** statistical analysis is right, but the other two are wrong, results are still unbiased

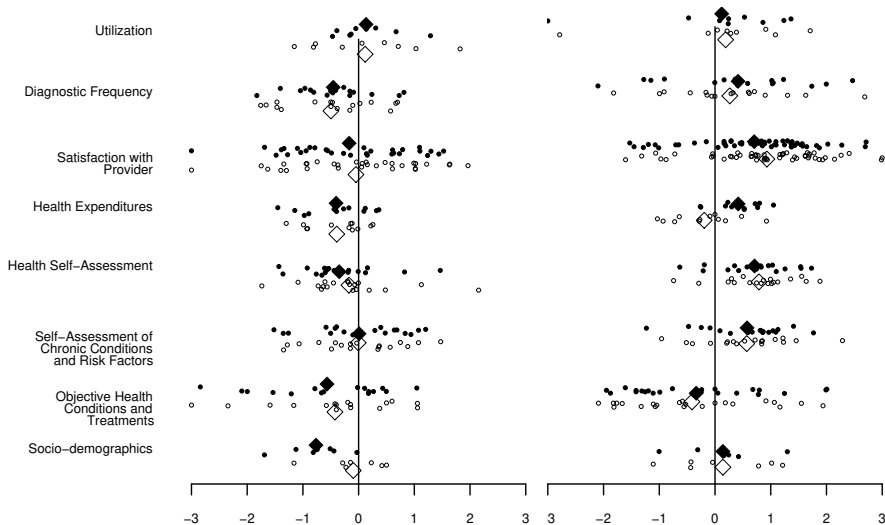
Two Additional Checks if Triple Robustness Fails

- 1 If one of the three works, then “effect of SP” on time 0 outcomes (measured in baseline survey) must be zero
- 2 If we lose pairs, we check for selection bias by rerunning this check

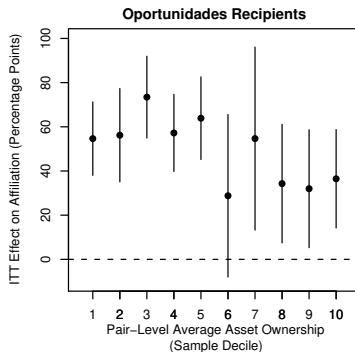
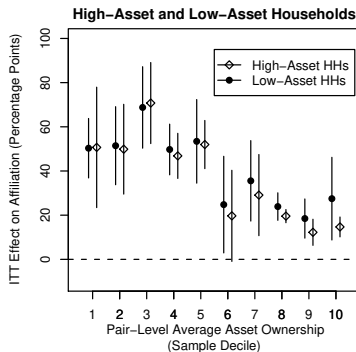
ITT on Outcome Measures at Baseline, for all families (left) and poor families, in Oportunidades (right)



ITT on Outcome Measures at Baseline, for wealthy families (left) and middle income families (right)



Effect of Encouragement on Seguro Popular Affiliation



Horizontal axes: per-capita asset ownership deciles of areas (poorer to the left). Vertical axes: percentage point causal effect of encouragement to affiliate on Seguro Popular affiliation.

Poor areas, not poor households, are affiliated the most

Effect on % of Households with Catastrophic Health Expenditures

	All Study Participants			Experimental Compliers		
	Average (Control)	ITT	SE	Average (Control)	CACE	SE
All	8.4	1.9*	(.9)	9.5	5.2*	(2.3)
Low Asset	9.9	3.0*	(1.3)	11.0	6.5*	(2.5)
High Asset	7.1	0.9	(0.8)	7.9	3.0	(2.7)
Female-Headed	8.5	1.4	(1.1)	10.6	3.8	(3.0)

“Catastrophic expenditures”: out-of-pocket health expenses $> 30\%$ of post-subsistence income

Effect on Out-of-pocket Health Expenditures, I (in pesos)

	All Study Participants			Experimental Compliers		
	Average (Control)	ITT	SE	Average (Control)	CACE	SE
Overall:						
All	\$1631.3	\$258.0	(\$175)	\$1712.7	\$689.7	(\$453)
Low Asset	1360.2	425.6*	(197)	1502.6	915.3*	(392)
High Asset	1867.9	128.4	(201)	1933.2	428.2	(669)
Female-Headed	1509.1	156.5	(207)	1689.9	428.6	(566)
Inpatient Care:						
All	532.5	96.9*	(44)	557.1	259.1*	(112)
Low Asset	527.1	188.2*	(73)	579.0	404.8*	(142)
High Asset	537.2	31.1	(52)	536.2	103.6	(173)
Female-Headed	452.5	115.1*	(68)	510.0	315.2*	(182)
Outpatient Care:						
All	448.3	116.7*	(63)	499.1	312.0*	(161)
Low Asset	412.3	176.7*	(73)	466.3	380.0*	(147)
High Asset	479.7	81.9	(69)	533.0	272.9	(230)
Female-Headed	416.3	110.4	(75)	496.8	302.4	(202)

Effect on Out-of-pocket Health Expenditures, II (in pesos)

	All Study Participants			Experimental Compliers		
	Average (Control)	ITT	SE	Average (Control)	CACE	SE
Medicine:						
All	521.1	20.0	(41)	534.5	53.3	(109)
Low Asset	427.3	17.8	(46)	444.7	38.3	(100)
High Asset	603.0	29.4	(47)	627.5	98.1	(157)
Female-Headed	625.6	53.6	(55)	738.9	146.8	(151)
Medical Devices:						
All	139.7	-8.8	(23)	117.8	-23.4	(62)
Low Asset	72.0	-0.2	(20)	72.8	-0.5	(43)
High Asset	198.8	-16.5	(29)	165.6	-55.1	(98)
Female-Headed	155.5	10.9	(34)	162.8	30.0	(94)

Utilization: Overall

	All Study Participants			Experimental Compliers		
	Average (Control)	ITT	SE	Average (Control)	CACE	SE
Utilization (Procedures):						
Used Outpatient Services (%)	62.6	−1.5	(1.9)	64.8	−4.0	(5.2)
Outpatient Visits (count)	1.6	−0.03	(0.09)	1.7	−0.08	(0.23)
Hospitalized (%)	7.6	−0.2	(0.5)	7.9	−0.5	(1.5)
Hospitalizations (count)	0.1	−0.003	(0.006)	0.1	−0.01	(0.02)
Satisfaction with Provider (%)	68.0	−1.0	(1.6)	69.8	−2.6	(4.5)
Utilization (Preventative) (%):						
Eye Exam Last Yr.	10.0	−0.7	(0.7)	9.8	−1.8	(1.9)
Flu Vaccine	25.7	−1.8	(1.4)	27.2	−4.9	(3.7)
Mammogram Last Yr.	5.1	−0.9	(0.6)	5.2	−2.3	(1.6)
Cervical Last Yr.	21.8	−1.3	(2.0)	22.2	−3.2	(4.8)
Pap Test Last Yr.	31.9	−2.3	(2.1)	33.2	−5.8	(5.0)

Self-Assessment: Overall

	All Study Participants			Experimental Compliers		
	Average (Control)	ITT	SE	Average (Control)	CACE	SE
Overall Health	55.7	4.2*	(2.0)	54.3	8.9*	(3.9)
Mobility	86.7	1.0	(1.0)	86.3	2.1	(2.0)
Vigorous Activity	69.2	4.6*	(2.7)	67.9	9.8*	(5.7)
Self-Care	95.3	0.4	(0.6)	95.2	0.8	(1.2)
Soreness	80.3	2.6*	(1.5)	79.3	5.5*	(3.1)
Pain	82.4	2.4*	(1.4)	81.4	5.2*	(2.8)
Sleeping	85.1	2.7*	(1.3)	84.3	5.9*	(2.5)
Depression	77.3	6.4*	(3.7)	76.0	13.8*	(7.3)
Anxiety	85.9	3.1	(2.0)	85.2	6.7	(4.1)

Self-Assessment, Controlling for Baseline Levels

	ITT		CACE	
Overall Health	0.6	(2.2)	1.7	(6.0)
Mobility	0.2	(0.9)	0.6	(2.5)
Vigorous Activity	3.3	(2.4)	8.9	(6.4)
Self-Care	-0.2	(0.6)	-0.5	(1.6)
Soreness	1.0	(1.4)	2.6	(3.8)
Pain	1.1	(1.2)	3.0	(3.3)
Sleeping	1.0	(1.0)	2.6	(2.5)
Depression	0.6	(3.0)	1.5	(7.9)
Anxiety	0.8	(1.8)	2.1	(4.8)

A **difference-in-difference test**: The causal effect of Seguro Popular on the change from baseline to followup in the difference between treated and control groups on health self-assessment variables.

Conclusions

Conclusions

- Positive effects detected now:

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- Positive effects not yet seen:

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- Positive effects not yet seen:
 - Expenditures on medicines

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- Positive effects not yet seen:
 - Expenditures on medicines
 - Utilization (preventative and procedures)

Conclusions

- **Positive effects detected now:**
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- **Positive effects not yet seen:**
 - Expenditures on medicines
 - Utilization (preventative and procedures)
 - Risk factors

Conclusions

- Positive effects detected now:
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- Positive effects not yet seen:
 - Expenditures on medicines
 - Utilization (preventative and procedures)
 - Risk factors
- Other findings:

Conclusions

- **Positive effects detected now:**
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- **Positive effects not yet seen:**
 - Expenditures on medicines
 - Utilization (preventative and procedures)
 - Risk factors
- **Other findings:**
 - Only 66% of automatically affiliated Oportunidades respondents were aware of this fact

Conclusions

- **Positive effects detected now:**
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- **Positive effects not yet seen:**
 - Expenditures on medicines
 - Utilization (preventative and procedures)
 - Risk factors
- **Other findings:**
 - Only 66% of automatically affiliated Oportunidades respondents were aware of this fact
 - More encouragement to affiliate might be devoted to finding the poor hidden within relatively “wealthier” clusters

Conclusions

- **Positive effects detected now:**
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- **Positive effects not yet seen:**
 - Expenditures on medicines
 - Utilization (preventative and procedures)
 - Risk factors
- **Other findings:**
 - Only 66% of automatically affiliated Oportunidades respondents were aware of this fact
 - More encouragement to affiliate might be devoted to finding the poor hidden within relatively “wealthier” clusters
 - Developed new and more powerful evaluation design and statistical methods, tuned to the needs of Mexico

Conclusions

- **Positive effects detected now:**
 - Catastrophic expenditures slashed
 - In-patient out-of-pocket expenditures drastically reduced
 - Out-patient out-of-pocket expenditures drastically reduced
 - Citizen satisfaction is high
- **Positive effects not yet seen:**
 - Expenditures on medicines
 - Utilization (preventative and procedures)
 - Risk factors
- **Other findings:**
 - Only 66% of automatically affiliated Oportunidades respondents were aware of this fact
 - More encouragement to affiliate might be devoted to finding the poor hidden within relatively “wealthier” clusters
 - Developed new and more powerful evaluation design and statistical methods, tuned to the needs of Mexico
 - Seguro Popular evaluation design: being copied around the world

For more information

<http://GKing.Harvard.edu>